Chapter 14: Collision Theory

November 2020

Introduction

As the name implies, collision theory involves objects, usually defined as a projectile or target, colliding with each other and how they scatter. Perhaps the most intuitive and familiar example would be a game of billiards. On a physically large scale such as this, it is easy to predict how the ball will scatter as we know the physical properties of the cue and target balls. Observing how they scatter won't tell us anything new about them. However, on a much smaller scale at the atomic and subatomic level where we can't directly measure the size or shape of particles, observing how projectiles scatter off of them can gives us information about their physical makeup. The most famous such application was when Ernest Rutherford used collision theory to discover the structure of the atom. By firing a stream of α particles at a thin layer of gold atoms on a gold foil and observing how they scattered, he was able to discover that most of the atomic mass was centered around a nucleus. Many more discoveries about atoms such as fusion or quarks have been made with the help of collision theory. At these sizes quantum physics is used, resulting in quantum collision theory being the most used type of collision theory. However, the fundamentals remain the same with classical collision theory. Therefore, becoming knowledgeable about classical collision theory provides a good introduction even for the quantum level.

I Scattering Angle and Impact Parameter

The **scattering angle** θ is the angle between the velocity of the incoming projectile before the collision and the outgoing velocity of the projectile after the collision. For a contact collision, the incoming and outgoing velocities could be measured immediately before and after the collision. For a scattering from a force field, they would need to be measured when the projectile is far enough away so the force is negligible. If the projectile does not collide with anything, then $\theta = 0$ meaning there was no scattering. If the collision is perfectly head on, then $\theta = \pi$.

The **impact parameter** b is the perpendicular distance of the incoming straight-line path of the projectile and the parallel axis going through the target's center, as shown in figure 1. The impact parameter essentially tells us how close the projectile was aimed at the target. Typically, the larger impact parameter corresponds to a smaller scattering angle, and likewise smaller impact parameter means larger scattering angle. With a sufficiently large impact parameter, its obvious θ will always be 0. Then the task of collision theory is to find a functional relation $\theta = \theta(b)$. Of course, physical properties of the target and projectile can impact☺the relationship, such as the malleability of the objects or if the projectile barrels straight though the target. In the context of trying to understand the basic concepts, we are assuming the objects are perfectly hard and spherical.

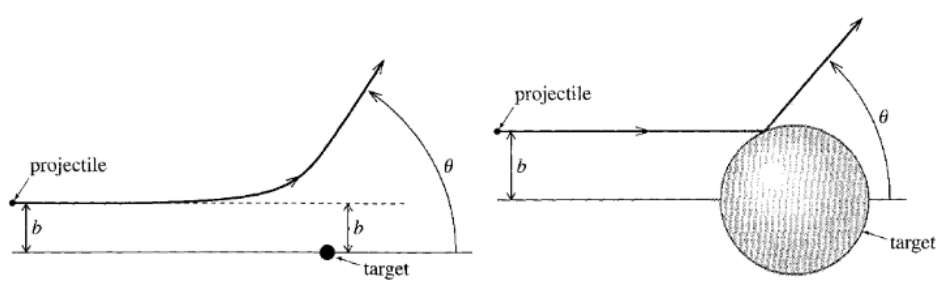

Figure 1: Taylor 14.1/14.2: Visual of the impact parameter b and scattering angle θ for force and contact collisions.

II The Collision Cross Section

If we assume that the projectile is moving in a perfectly straight line towards the target or targets, then only the the largest perpendicular cross sectional area of the target relative to projectile's velocity is important in determining whether there will be a collision. Just like a light source directly in front of a thin disk casts the same shadow as a long cylinder. This cross section is denoted σ so with spherical targets $\sigma = \pi R^2$.

Suppose we have a multiple targets within a certain area, called the target assembly. If we operate on the assumption that none of the targets overlap as seen from the direction of the projectile, and we know the target density n_{tar}, then it is easy to calculate the total area of the targets. Letting A be the total area of the target assembly, then $n_{tar}A\sigma$ gives us the total area of all the targets. Something to keep in mind is that n_{tar} is not simply the number of targets, it is specifically the target *density*, or number of targets per area. So $n_{tar}A$ is the number of targets. This isn't very useful on the large scale with few objects, but is useful if the target area is very dense or for small scales like in the molecular level.

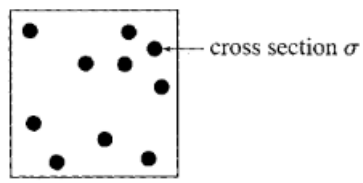

Figure 2: Taylor 14.4: A target assembly of spheres as seen from the perspective of a projectile.

The probability of any given projectile colliding with a target is as follows:

$$(\text{Probability of hit}) = \frac{\text{area occupied by targets}}{\text{total area}} = \frac{n_{tar}A\sigma}{A} = n_{tar}\sigma$$

If we send a large number of projectiles through the target assembly, then if N_{inc} is the number of incident projectiles and N_{sc} is the number of projectiles scattered, then it is simple to calculate

$$N_{sc} = N_{inc}n_{tar}\sigma.$$

In practice, the projectile won't be a point particle. If we take the projectile to be another hard sphere with radius R_1, and the target with radius R_2, then there is a collision if the the impact parameter $b \leq R_1 + R_2$. Then if still consider the center of the projectile as a point particle, the effective scattering cross section σ now becomes $\sigma = \pi(R_1 + R_2)^2$, and our previous equations remain the same.

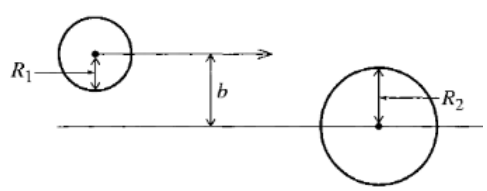

Figure 3: Taylor 14.5: Both target and projectile are spheres. Collision occurs if $b \leq R_1 + R2$

If we consider objects that are not perfectly hard, is possible that different sections of the target interact differently with the projectile, such as if certain sections were putty like and cause the projectile to stick. Then we could follow the same process to find a σ that corresponds to the cross section of this sticky section. At the atomic level, there could be ionization cross sections and fission cross sections for where an electron hits an atom. since we are specifically interested in the collision cross section that causes scattering, we will just use σ to refer to scattering cross section. In other cases, we would subscript σ based on what cross section we were interested in.

III The Differential Scattering Cross Section

While finding σ we just considered whether the projectiles were scattered at all. In order to monitor which direction the projectiles get scattered in, we need to do a bit more work.

Solid Angle

The standard way of measuring the direction is to make the z axis incident to the direction the projectile is traveling in, and then use polar angles (θ, ϕ) for the direction of the scatter, with θ coming down from the z axis ranging from $0 \to \pi$, and ϕ rotating around from $0 \to 2\pi$. since there are infinitely many angles in this range, we cannot calculate how many particles will scatter in exactly a given angle (θ, ϕ). Instead, we must consider some narrow cone around (θ, ϕ). To characterize this cone, we must use the notion of a solid angle.

The solid angle is similar to the ordinary angle of a circle. If s is some arc length of a circle of radius r, then the ordinary angle $d\theta = \frac{s}{r}$. For a sphere of three dimensions, we use some small area of the surface proportional to r^2 so the **solid angle** $d\Omega = \frac{A}{r^2}$.The ordinary angle ranges on a circle to 2π radians, while the solid angle works in three dimensions and ranges to 4π of a unit called steradians. The area A works for any shape of cone, so we can choose to find the area of a "rectangular" section using spherical polar coordinates. Then this area is $r^2 sin(\theta) d\theta d\phi$ and the solid angle is

$$d\Omega = sin(\theta) d\theta d\phi$$

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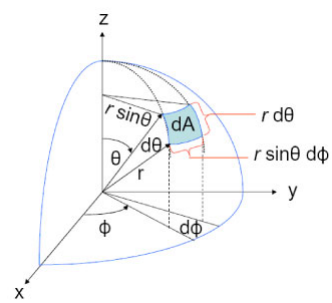

Figure 4: Visualisation of finding a section of surface area in spherical polar coordinates.

Differential Cross Section

If we let $N_{sc}(\text{into } d\Omega)$ represent the number of projectiles scattered into some solid angle $d\Omega$ of some direction (θ, ϕ), then we can use our familiar argument of

$$N_{sc}(\text{into } d\Omega) = N_{inc} n_{tar} d\sigma(\text{into } d\Omega)$$

where $d\sigma(\text{into } d\Omega)$ is the effective cross section for scattering into the solid angle $d\Omega$. Our formula for $d\sigma(\text{into } d\Omega)$ is

$$d\sigma(\text{into } d\Omega) = \frac{d\sigma}{d\Omega} d\Omega$$

where $\frac{d\sigma}{d\Omega}$ is the **differential scattering cross section**. If we add all the $N_{sc}(\text{into } d\Omega)$ for all possible $d\Omega$, then we get our total N_{sc} for all scatterings. In other words, we just integrate over all angles to get $N_{sc} = N_{inc} n_{tar} \sigma$. Then to get the total scattering cross section

$$\sigma_{sc} = \int \frac{d\sigma}{d\Omega} d\Omega$$

Calculating the differential cross section

Lets continue our example of hard sphere scattering and calculate the differential cross section. By calculating a projectiles trajectory, we can solve for $\theta = \theta(b)$ and likewise $b = b(\theta)$. Since a sphere is axially symmetric, our calculation can include all values of ϕ at the same time.

Consider all possible projectiles approaching a spherical target with impact parameters between b and $b + db$. Then since we can include all 2π of ϕ, the impact parameters are incident on an annulus with

$$d\sigma = 2\pi b db$$

where the $2\pi b$ term comes from the circumference of the annulus and db is the width. Similarly, the projectile will scatter between θ and $\theta + d\theta$ in the solid angle

$$d\Omega = 2\pi sin\theta d\theta$$

where we used the solid angle formula and just plugged in 2π for $d\phi$. Then $\frac{d\sigma}{d\Omega}$ is easily calculated by dividing the two equations, so

$$\frac{d\sigma}{d\Omega} = \frac{b}{sin\theta}\left|\frac{db}{d\theta}\right|$$

where the absolute value is used to ensure the differential cross section is positive.

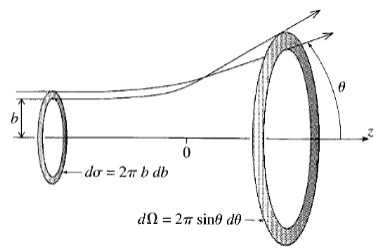

Figure 5: Taylor 14.9: Projectiles within impact parameter $b + db$ scattering into solid angle $d\Omega = 2\pi sin\theta db$.

By calculating b and θ we can solve for the total scattering cross section. Using the law of reflection, define α as shown in the figure below.

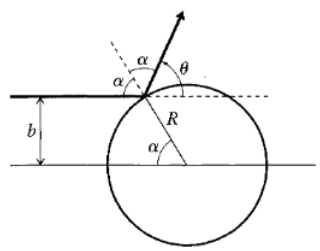

Figure 6: Taylor 14.10

We have $b = Rsin\alpha$ and $\theta = \pi - 2\alpha$. Then $\alpha = \frac{\pi-\theta}{2}$. So $b = Rsin(\frac{\pi-\theta}{2}) = Rcos(\frac{\theta}{2})$, and now we have $\left|\frac{db}{d\theta}\right| = \frac{R}{2}sin(\frac{\theta}{2})$. Using our differential cross section equation

$$\frac{d\sigma}{d\Omega} = \frac{Rcos(\frac{\theta}{2})}{sin\theta}\frac{Rsin(\frac{\theta}{2})}{2} = \frac{R^2}{4}$$

where we used the identity $cos(x)sin(x) = \frac{sin(2x)}{2}$. Then

$$\sigma = \int \frac{R^2}{4}d\Omega$$

and integrating over all 4π solid angles we have $\sigma = \pi R^2$ which is the total scattering cross section of a sphere.

IV Example Problem

Instead of spheres, we will imagine we have pucks on an ice rink, so the scattering will essentially be in two dimensions as we consider the surface of the ice rink to be a plane. Then their cross section is just their width or diameter.

a) Show that in 2D

$$\frac{d\sigma}{d\theta} = \left|\frac{db}{d\theta}\right|$$

Consider a projectiles approaching a target between impact parameters b and $b + db$. In our previous 3D example, we included a $2\pi b$ term. Since we don't have ϕ in 2D, particles will only get scattered in the $b + db$ strip. Then the change in the scattering cross section is only db so $d\sigma = db$. Instead of the solid angle, in a 2D circle we just need to solve for the ordinary angle $d\theta = \frac{s}{r}$, where s is the arc length defined as $s = rd\theta$. Then we get the obvious answer $d\theta = d\theta$. Therefore $\frac{d\sigma}{d\theta} = \left|\frac{db}{d\theta}\right|$.

b) Find the differential cross section of the scattering of a small projectile off a hard disk of radius R

The scattering off a disk looks the same as figure 6, and therefore our equations for b and θ are the same where $b = Rcos(\frac{\theta}{2})$. Then

$$\frac{d\sigma}{d\theta} = \left|\frac{db}{d\theta}\right| = \frac{Rsin(\frac{\theta}{2})}{2}$$

c) Integrate to find total cross section

Integrating with θ from $0 \to 2\pi$ we have

$$\sigma = \int \frac{d\sigma}{d\theta} d\theta = \int_0^{2\pi} \frac{Rsin(\frac{\theta}{2})}{2} d\theta = \frac{R}{2}\int_0^{2\pi} sin(\frac{\theta}{2}) d\theta = \frac{R}{2}4 = 2R$$

which is the answer we want as the scattering cross section of a disk is its diameter or $2R$.

d) Now suppose the projectile is another hard disk of radius $2R$ and the targets are hard disks of radius R pinned to the ice

i What is the differential cross section now?

Because the projectile has a non-negligible radius and the impact parameter measures from the center of the target to the center of the projectile, we need to add its radius to the radius of the target. So the impact parameter is now

$$b = (R + 2R)sin(\frac{\theta}{2}) = 3Rsin(\frac{\theta}{2})$$

and our differential cross section is $\frac{3Rsin(\frac{\theta}{2})}{2}$ where integrating yields $\sigma = 6R$ which is what we expect.

ii. Suppose there are 2000 pucks of radius $R = 3cm$ in a square kilometer of ice such that none of them overlap from the side, and you randomly hit pucks of radius $2R$ into the distribution, what fraction of pucks would you expect to make it through without scattering?

As we calculated, $\sigma = 6R = .18m$. The area as seen from the side is just the length of the target assembly, so $1000m$. We have 2000 targets that don't overlap from the side, so the target density is $\frac{2000}{1000m} = 2$ per m. Then (Probability of hit) = 2(.18) = .36. Then the fraction of pucks that make it through without scattering is 1 − .36 = .64.

iii. Test your value by writing a program to simulate the above scenario

Imagine the ice rink surface an being the xy plane, and along the x axis is where we line up our shots. Then as we shoot them parallel to the y axis, a collision occurs only if a shot shares a same x value as a target. Since the y value, or depth, doesn't matter a seen from the side, we only need the program to care about the x values.

```
import numpy as np
import random
```

```
R=0.03 #radius R
shotarray=np.arange(2*R,1000-2*R,.005)#possible locations of shooting pucks

el=np.arange(R,1000-R,2*R+.001) #possible locations of target pucks
stuckpucks=[]
for l in el:
     stuckpucks.append(l) #storing locations into a list
```

We begin by initializing a list of possible x values, or locations, for where the pucks will be hit from. The locations are based on the center the pucks and we don't want part of the pucks to go past the edge of the assembly, so the limits are one radius away from each edge. Recall the radius of these pucks is $2R$. There are no restrictions on overlapping of these pucks so the interval .005 was chosen to cover as many possible locations in the range since we can't store all possible values from $0 \to 1000$ continuously. We repeated this for possible locations of the target pucks, but since we did have an overlap restriction, the intervals were set to $2R$ with a small .001 separation so adjacent targets don't overlap. While this doesn't make the distribution completely random as we divided the range into discrete locations, we are still left with over 16,600 possible locations for 2,000 targets which should still be enough randomness for this simple scenario. The target locations were appended to a separate list so we could use random.sample() later in the program.

```python
def sim():  #program to simulate shooting pucks at targets
    shots=[]
    for i in range(2000): #taking 2000 shots and choosing random values from the possible locations, with replacement
        shot=random.choice(shotarray)
        shots.append(shot)

    pucks=random.sample(stuckpucks,2000) #seleceting random values from the possible target locations without replacement

    hits=0 #initializing hits variable
    for shot in shots:   #nested loop to check if each shot hits any of the targets
        for puck in pucks:
            b=abs(shot-puck) #calculating the impact paramter for each shot and target
            if b <= 3*R:
                hits+=1  #adding 1 to hits if impact parameter is less than or eqaul to 3R
    phit=hits/len(shots)  #fraction of shots that hit a target
    pmiss=1-phit   #fraction of shots that didn't scatter

    return pmiss #retrun the fraction that didn't scatter
```

We now defined the function to run the simulation. Since the shots can overlap, we chose locations randomly with replacement from the possible locations list. We chose to take 2000 shots so we looped through 2000 times of selecting a location and appending it to the shots list. For the target pucks, we needed to select values from the possible locations list but use replacement this time. This is easily done with random.sample(). Then we used a nested loop to check if the impact for each of the shot and target pucks was less than $3R$, in which case we added 1 to the hits counter. Then we divide the number of hits by the number of shots to get our fraction of shots that hit, subtracted it from 1 to get how many didn't scatter, and returned our value.

```python
sim() #running the program
```

```
0.6575
```

```python
results=[] #storing values of multiple runs
for i in range(25):
    results.append(sim())
```

```python
np.mean(results) #taking the average of all the values to get the most accuarate probability
```

```
0.64058
```

Running the program once we see our value is close, but noticeably larger than our .64 calculation. This isn't unexpected as the locations were random, so there could be a much less dense section of targets where a larger number of shots were taken resulting in more making it through. If we run the program multiple times and average all the values, then we end up with our calculated result.